

01 What am I attempting?

I want to run the agricultural damage data backwards in time to reconstruct what Gaza’s agricultural landscape may have looked like before colonial occupation. Starting from the vegetation health and density recorded in the dataset, I want to explore how the extent of agricultural land, vegetation density, and land-use patterns might change as we go backward in time to pre-colonial conditions.

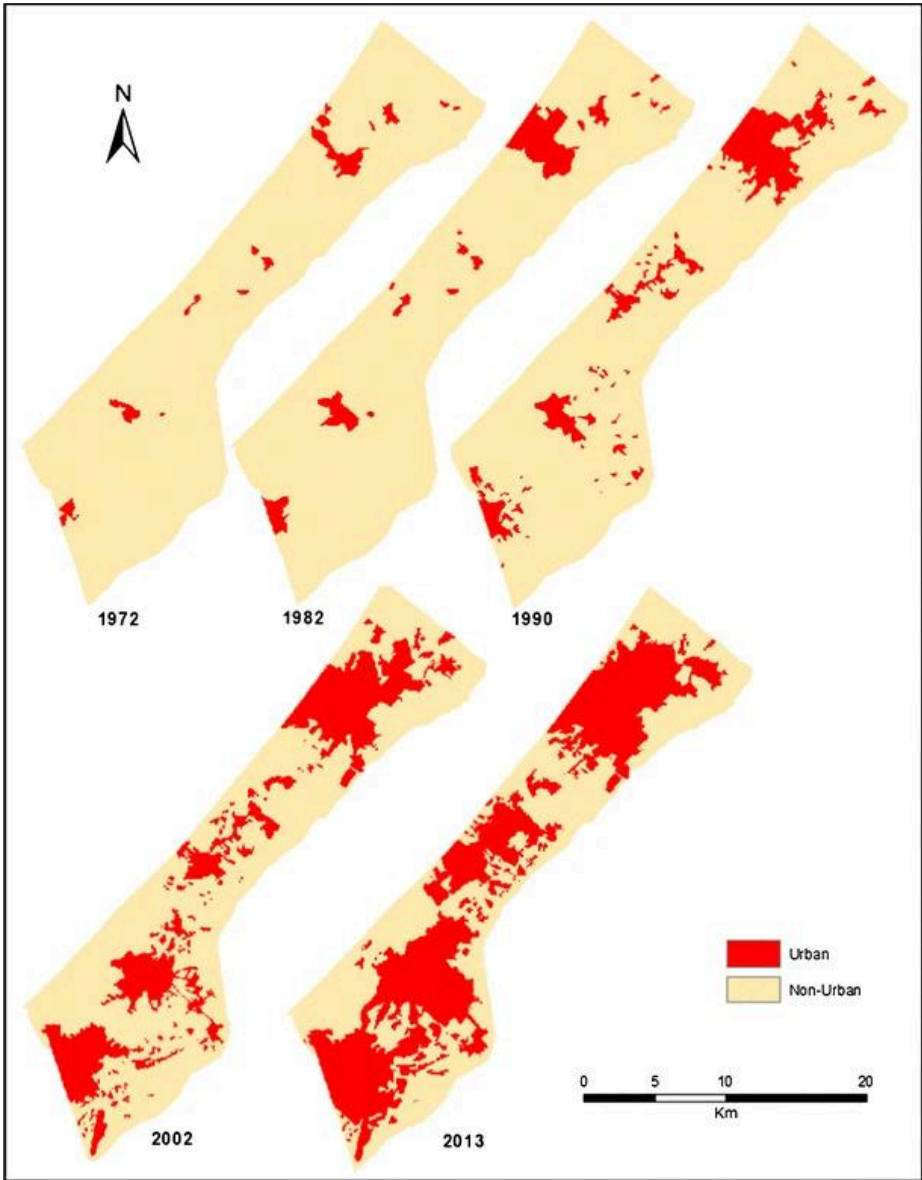
02 The Dataset (Spatial)

- Made by: UNOSAT (<https://data.humdata.org/dataset/unosat-gaza-strip-agricultural-damage-assessment-november-2023>)
- Date: 17 November 2023 assessment
- Purpose: assess damage to agricultural land in the Gaza Strip using Sentinel-2 satellite imagery, comparing 2023 vegetation conditions against 2017–2022 (acts only as a source material baseline, no data).
- Method: changes in vegetation health/density were identified through satellite imagery by comparing them.

Date	Damaged	Non-affected
27 Oct 2023	11.76 km ²	171.21 km ²
11 Nov 2023	24.01 km ²	158.25 km ²

03 Research

I found existing research (*Scenario of land use and land cover change in the Gaza Strip using remote sensing and GIS models*) that uses remote sensing to reconstruct and model changes to Gaza's landscape through time. It uses historical Landsat imagery from 1972, 1982, 1990, 2002 and 2013 to classify land use and analyze transitions between different landscape conditions. It provides historical spatial records against which a reconstruction could be developed.



04 My Record

My starting record would be the 11 November 2023 UNOSAT assessment. It contains two classifications: damaged and non-affected agricultural land—with their area in km² and polygon geometries showing their spatial distribution across Gaza.

2023 → 2013 → 2003 → 1993 → 1983....

05 My Rule

I am adapting the vegetation recovery model developed by Gouveia, DaCamara and Trigo in 2010 (<https://nhess.copernicus.org/articles/10/673/2010/>), which uses NDVI to measure how vegetation greenness changes following disturbance.

The model first defines a healthy reference condition for vegetation and calculates the difference between the observed NDVI and this reference, called the “lack of greenness.”

This difference is modeled using an exponential curve $y(t) = a e^{-bt}$
a – represents the initial loss of greenness
t – time
b – the vegetation recovery rate

For my run, I would apply this logic backward: at each timestep, the NDVI of damaged agricultural vegetation increases toward its healthy reference condition. I am uncertain about this because I would be using the source material and making abstractions (?).

06 My Run

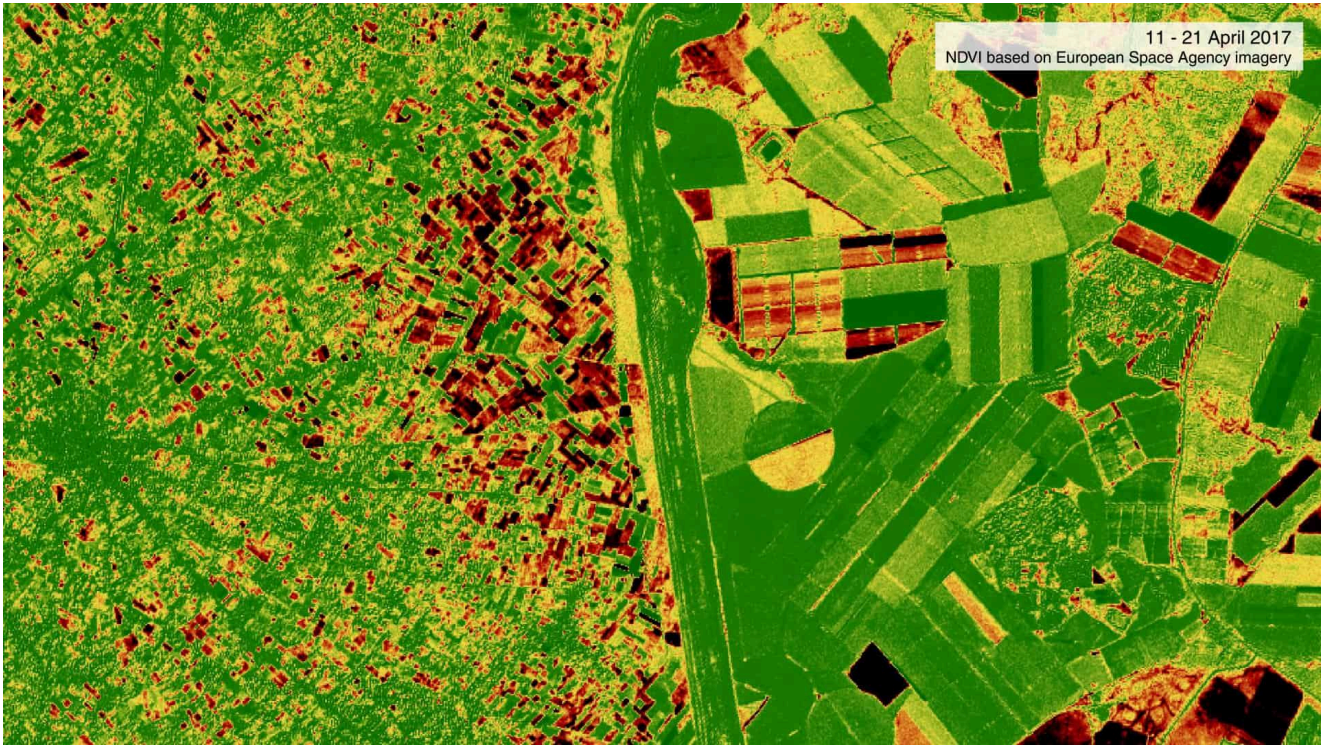
The simulation runs backward from 2023 in 10-year intervals (2023 → 2013 → 2003 → 1993...). At each step, the NDVI rule is applied to estimate an earlier vegetation condition. The run is deterministic.

07 Data it would generate

Date	Location	Starting NDVI	Modeled NDVI	Change
2023	Polygon A	0.XX	0.XX	+X
2013	Polygon A		0.XX	+X
2003	Polygon A		0.XX	+X
1993	Polygon A		0.XX	+X
1983	Polygon A		0.XX	+X

08 What it can't see

- Specific military events, bombing, bulldozing, fires, or herbicide use
- Destruction of other ecologies like irrigation and water infrastructure
- Crop type, farming practices, or differences between plant species
- Soil degradation, contamination, salinity, and water availability
- Climate change, drought, rainfall, and extreme heat
- Displacement, restrictions on land access, and changing political borders
- Colonial land policies and historical systems of land control
- Actual vegetation conditions before satellite records, we get more and more speculative



Molavi, Shorideh. <https://forensic-architecture.org/investigation/herbicidal-warfare-in-gaza>